

Section A — Concept Application

Q-1.

Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

1: Choose the motherboard:

Select a motherboard based on the required CPU socket, such as AM5 or LGA1700. Also check that it supports the required RAM type and has the necessary storage slots, such as M.2 or SATA.

2: Choose the CPU:

Pick a CPU that uses the same socket as the motherboard. Check the motherboard manufacturer's CPU compatibility list to confirm that the processor is supported. Also check BIOS requirements.

3: Choose the RAM:

Select RAM that the motherboard supports, such as DDR4 or DDR5. Check the maximum supported capacity and RAM speed. The RAM type must match the motherboard because DDR4 and DDR5 are not interchangeable.

4: Choose the storage device:

For an office workstation, an NVMe M.2 SSD is a good choice because it provides fast boot and application loading. Confirm that the motherboard has a compatible M.2 slot. SATA SSDs can also be used if the motherboard has SATA ports.

5:Confirm compatibility:

Before purchasing, compare the specifications of all four components. Check CPU socket, motherboard CPU support, RAM type/speed, M.2 or SATA interface, physical dimensions, and BIOS support. The motherboard manufacturer's compatibility list and supplier specifications can be used to verify that all parts will work together.

Q-2.

Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

Feature	HDD	SSD (SATA)	NVMe SSD
Technology	Magnetic spinning disks	NAND flash memory	NAND flash using PCIe
Speed	Slowest, ~80–160 MB/s	Around ~500–550 MB/s	Usually ~3,500–7,000+ MB/s
Moving parts	Yes	No	No
Durability	Lower	Higher	Higher
Noise	Produces noise	Silent	Silent
Power use	Higher	Lower	Generally low
Cost per GB	Cheapest	Moderate	Higher than HDD

1. SSD is much faster than HDD, so the computer starts quickly and programs open faster.
2. SSDs have no moving parts, so they are quieter and more resistant to damage.
3. NVMe SSDs are even faster, but normal office work usually does not need that extra speed.

4. A SATA SSD gives a good balance of speed, reliability, storage, and cost.
5. An HDD is useful when you need a lot of storage at a low price.

Q-3.

Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.

GPU and Cooling System

The GPU (Graphics Processing Unit) is responsible for handling graphics and visual tasks in design software. It helps render images, 3D models, animations, and other graphics smoothly. A good GPU can make the software faster and more responsive, especially when working with large or complex designs.

The cooling system keeps the GPU and other components at a safe temperature during long hours of use. It uses fans, heatsinks, and sometimes liquid cooling to remove heat. If the system gets too hot, the GPU may slow down automatically or the computer may become unstable or shut down.

Conclusion: A suitable GPU provides good graphics performance, while an effective cooling system prevents overheating. Together, they help the workstation remain fast, stable, and reliable during several hours of continuous work.

Q-4.

Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.

Hardware compatibility is important because all PC components must work together correctly. The CPU must match the motherboard socket, the RAM must be supported by the motherboard, and the power supply must provide enough power and the correct connectors. Storage devices and graphics cards must also use supported interfaces such as SATA, M.2, or PCIe.

If incompatible components are used, problems may include:

1. PC does not start because the CPU or RAM is not supported.
2. RAM may not work if the type or speed is incompatible.
3. Storage may not be detected if the motherboard does not support the drive interface.
4. GPU may not fit or work properly with the motherboard or power supply.
5. Power problems can occur if the PSU does not provide enough wattage or the correct connectors.
6. Overheating may happen if the CPU cooler is incompatible or insufficient.
7. System crashes or poor performance can occur due to incompatible hardware or drivers.

Therefore, before buying components from different vendors, check the motherboard and CPU compatibility, RAM type, storage interface, PSU wattage/connectors, and physical dimensions. This helps ensure the computer is stable, safe, and works properly.

Q-5.

Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.

I would choose a PSU based on the total power requirements of all components. First, I would check the power needs of the CPU, GPU, motherboard, storage drives, and fans. Then I would choose a PSU with some extra capacity, usually 20–30% above the estimated maximum power, so it can handle peak loads safely.

For example, if the PC needs around 400 W, I would choose a good-quality 500–550 W PSU.

Before powering on, I would check:

1. Turn off and unplug the PSU before making connections.
2. Connect the 24-pin ATX cable to the motherboard.
3. Connect the 4/8-pin CPU power cable near the CPU socket.
4. Connect the required 6/8-pin PCIe power cables to the GPU.
5. Connect SATA power cables to SATA SSDs/HDDs.
6. Make sure every connector is fully inserted and correctly aligned.
7. Check that no loose cables can touch fans.
8. Make sure the PSU has enough wattage and correct connectors for the components.
9. Check for loose screws, metal objects, or damaged cables inside the case.
10. Ensure the motherboard is correctly mounted on its standoffs to prevent short circuits.

After these checks, I would connect the PSU to the wall and switch it on. Then I would perform the first boot and check that the system starts normally.

Q-6.

Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not. B
Section B — Practical Coding Task.

Checking POST Before Handover

POST (Power-On Self-Test) checks the basic hardware when the computer starts. I would perform these checks:

- 1.Power on the PC and check that the fans start spinning.
- 2.Confirm that the monitor displays the manufacturer's logo or BIOS/UEFI screen.
- 3.Check that there are no beep codes or warning lights from the motherboard.
- 4.Enter the BIOS/UEFI and confirm that the CPU, RAM, and storage drives are detected correctly.
- 5.Check that the correct amount of RAM is shown.
- 6.Confirm that the SSD/HDD is detected and available as a boot device.
- 7.Check that the CPU temperature is at a normal level in BIOS/UEFI.
- 8.Make sure the system continues from POST to the operating system boot screen.
- 9.Restart the PC several times to make sure it starts consistently.

If POST Fails

If the computer does not pass POST, I would:

- 1.Check that the PSU is switched on and power cables are connected correctly.

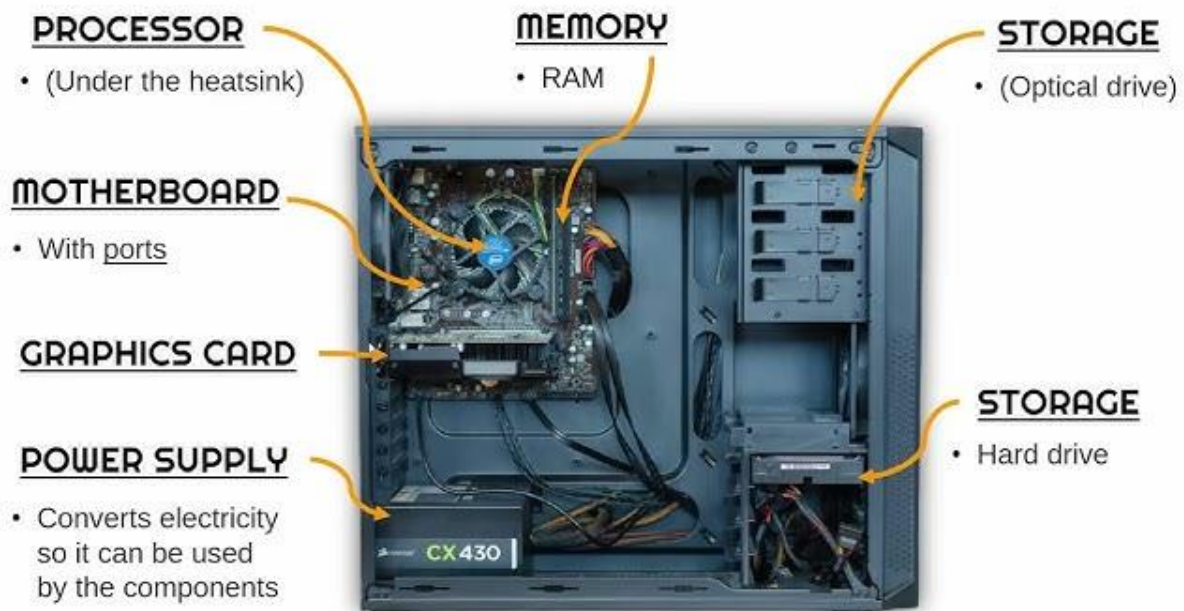
2. Reseat the RAM and try one RAM module at a time.
3. Check that the CPU power and motherboard 24-pin cables are connected.
4. Reseat the GPU and check its power cables if a dedicated GPU is installed.
5. Check the motherboard's debug LEDs or beep codes to identify the faulty component.
6. Disconnect unnecessary devices such as extra USB devices and storage drives.
7. Check for loose connections or incorrect motherboard standoffs.
8. Clear the CMOS/BIOS settings if necessary.
9. Test components one at a time to identify the faulty or incompatible part.

Once POST completes successfully and the operating system starts normally, I would carry out a final hardware check and then hand the workstation over to the client.

Section B — Practical Coding Tasks

Task 1: Component Identification & Functions:

1: Identify and describe the function of the motherboard, CPU, RAM, GPU, storage device, and cooling system:



Motherboard:

The main circuit board. It connects all computer components and allows them to communicate with each other.

CPU:

The brain of the computer. It processes instructions and performs calculations.

RAM:

Temporary memory that stores data and programs currently being used. More RAM helps the computer run multiple programs smoothly.

GPU:

Processes and displays graphics, images, and videos. It is especially important for gaming and graphic-design applications.

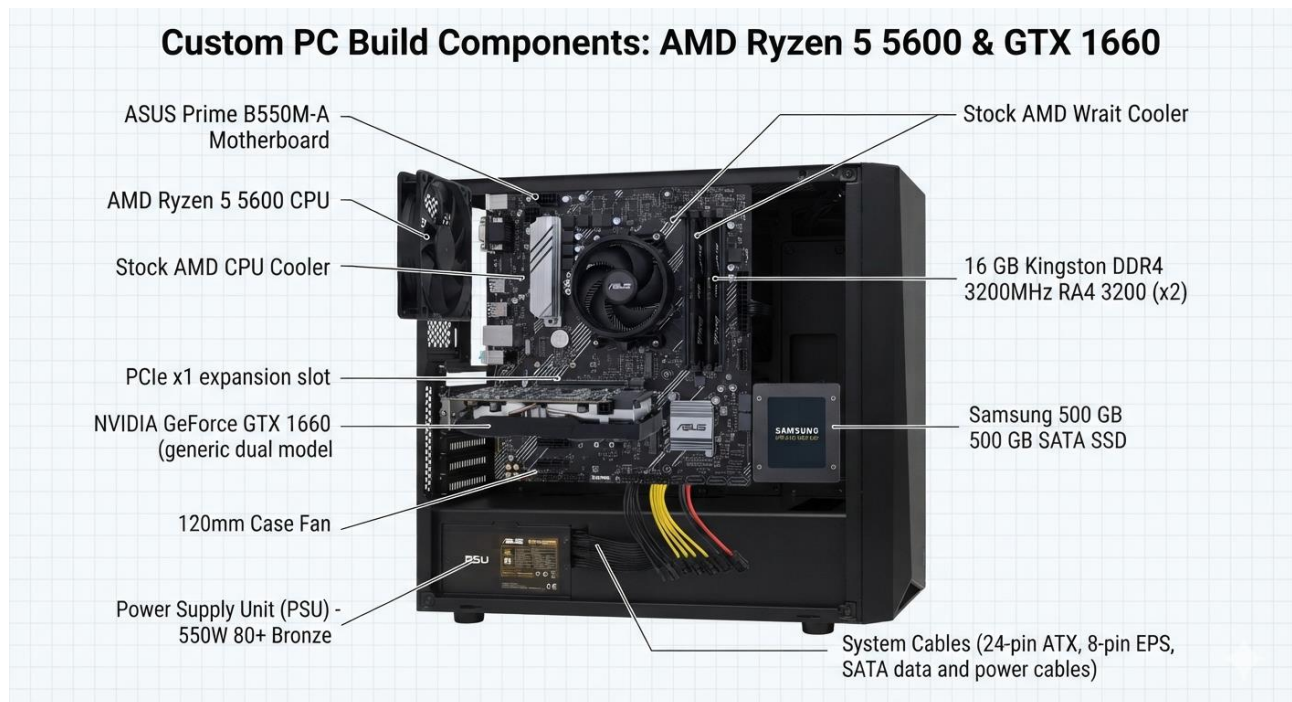
Storage Device:

Stores data permanently, such as the operating system, applications, documents, photos, and videos. Examples are HDD, SSD, and NVMe SSD.

Cooling System:

Removes heat from components such as the CPU and GPU. Fans, heatsinks, or liquid cooling help prevent overheating and maintain stable performance.

2:Note the make/model or key specification of each component where visible.

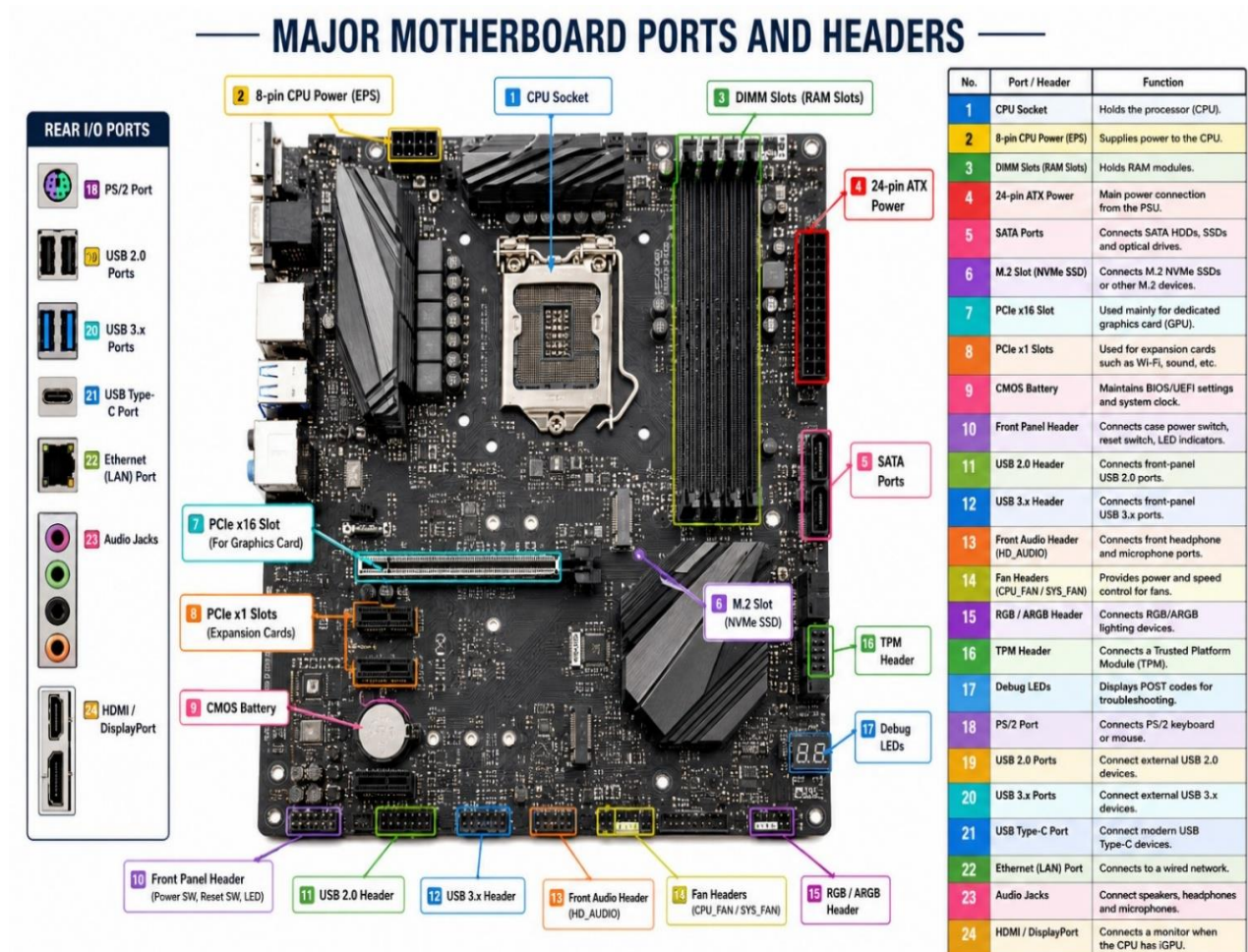


3: Present the findings as a labeled component list or table.

No.	Component	Make/Model or Key Specification	Function
1	Motherboard	ASUS Prime B550M-A / B550 chipset	Connects all computer components.
2	CPU	AMD Ryzen 5 5600 / 6 cores, 12 threads	Processes instructions and performs calculations.
3	RAM	Kingston 16 GB DDR4 3200 MHz	Temporarily stores data used by running programs.
4	GPU	NVIDIA GeForce GTX 1660 / 6 GB	Processes graphics, images, and video.
5	Storage Device	Samsung 500 GB SATA SSD	Permanently stores the operating system, applications, and files.
6	Cooling System	CPU heatsink + cooling fan	Removes heat and prevents overheating.

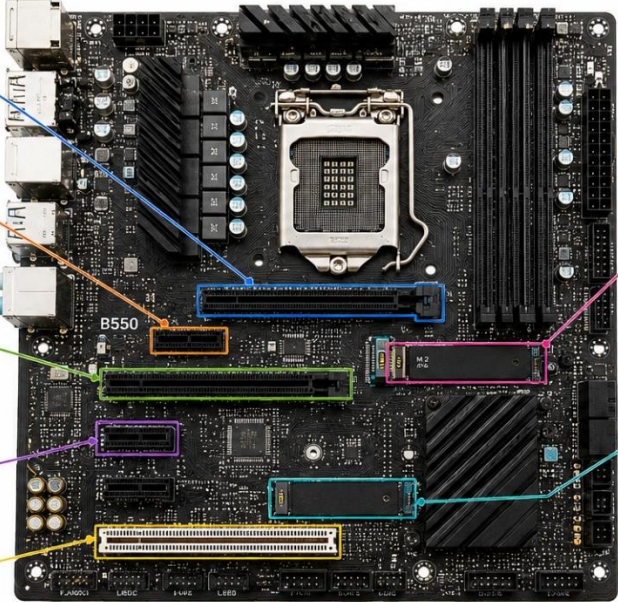
Task 2: Motherboard Ports & Socket Identification:

1: Identify and label all major motherboard ports (SATA, power, USB headers, etc.)



2. Identify all expansion slots (PCIe, M.2, etc.) and their purpose.

MOTHERBOARD EXPANSION SLOTS – SIMPLE GUIDE



1 PCIe x16 Slot
Main slot for graphics card (GPU).

2 PCIe x1 Slot
Used for small cards like Wi-Fi, sound card, network card, USB card, etc.

3 PCIe x4 Slot
Used for devices like NVMe adapter, capture card, network card, RAID controller, etc.

4 PCIe x1 Slot
Another x1 slot for expansion cards like Wi-Fi, sound card, etc.

5 PCI Slot (Legacy)
Old slot for legacy cards like sound, network, modem. Rare in modern motherboards.

6 M.2 Slot (Top)
For NVMe SSD (M.2 SSD). Super fast storage.

7 M.2 Slot (Bottom)
Another M.2 slot for NVMe SSD or SATA M.2 SSD (depends on the motherboard).

SLOT	PURPOSE
1	PCIe x16 Main slot for graphics card (GPU)
2	PCIe x1 For small cards (Wi-Fi, sound, network, USB card, etc.)
3	PCIe x4 For NVMe adapter, capture card, network card, RAID controller, etc.
4	PCIe x1 Another slot for expansion cards
5	PCI (Legacy) Old slot for legacy cards (rare now)
6	M.2 Slot (Top) For NVMe SSD (very fast storage)
7	M.2 Slot (Bottom) For NVMe or SATA M.2 SSD (depends on board)

QUICK NOTES

- PCIe x16 → Best for Graphics Card (GPU)
- PCIe x1 → Small cards like Wi-Fi, Sound, Network
- M.2 → Fast SSD (NVMe) for best performance
- The number and type of slots may vary with motherboard model.


Graphics Card
→ PCIe x16


Wi-Fi Card
→ PCIe x1


Sound Card
→ PCIe x1


NVMe SSD
→ M.2 Slot


Legacy Card
→ PCI Slot

PCIe x16 Slot : Mainly used for a dedicated graphics card (GPU). It provides high-speed communication between the GPU and motherboard.

PCIe x8 Slot : Used for high-bandwidth expansion cards, such as GPUs, network cards, or storage controllers.

PCIe x4 Slot : Used for devices such as NVMe adapters, network cards, sound cards, and storage controllers.

PCIe x1 Slot : Used for smaller expansion cards such as Wi-Fi/Bluetooth, sound cards, capture cards, and network cards.

M.2 Slot : Used mainly for NVMe SSDs. Some M.2 slots can also support SATA M.2 drives or Wi-Fi modules, depending on the motherboard.

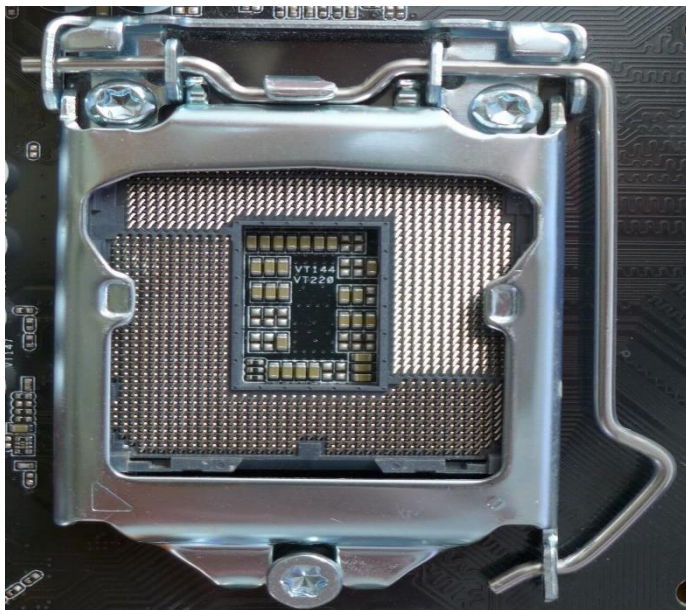
PCI (Legacy) : Older expansion slot used for devices such as sound, network, and modem cards. It is rarely found on modern motherboards.

M.2 E-Key Slot : Commonly used for Wi-Fi and Bluetooth modules on compatible motherboards.

DIMM Slots : Holds RAM modules. Technically these are memory slots rather than expansion slots, but they are important motherboard slots.

3. Identify the CPU socket type and note which CPU families are compatible with it.

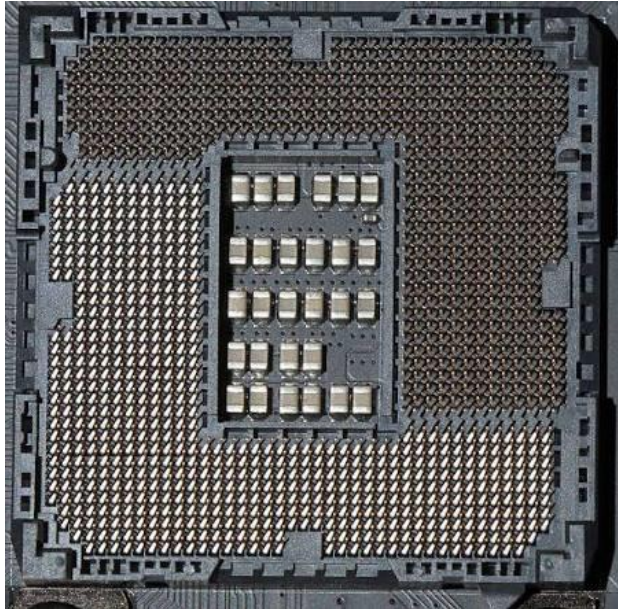
1. LGA 1151:



Compatible CPU Families :

Intel Core i3, i5, i7, i9 (6th–9th Gen, depending on chipset)

2. LGA 1200:



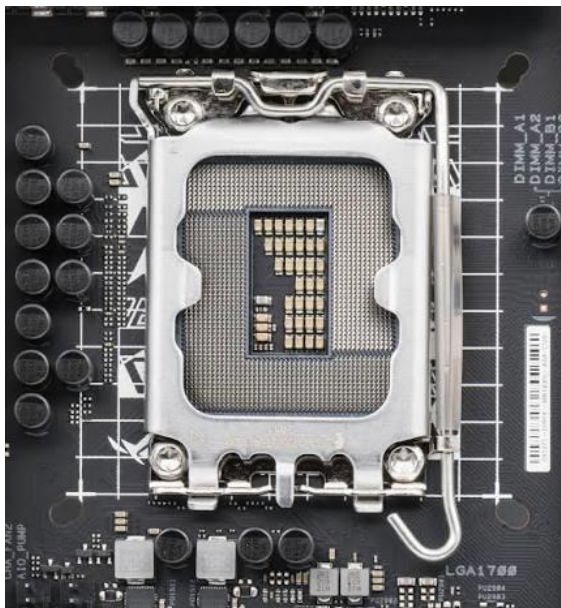
Compatible CPU Families :

Intel Core 10th and 11th Gen

3. LGA 1700:

Compatible CPU Families :

Intel Core 12th, 13th and 14th Gen



4. AM4 :



Compatible CPU Families :

AMD Ryzen 1000–5000 series (depending on motherboard/BIOS)

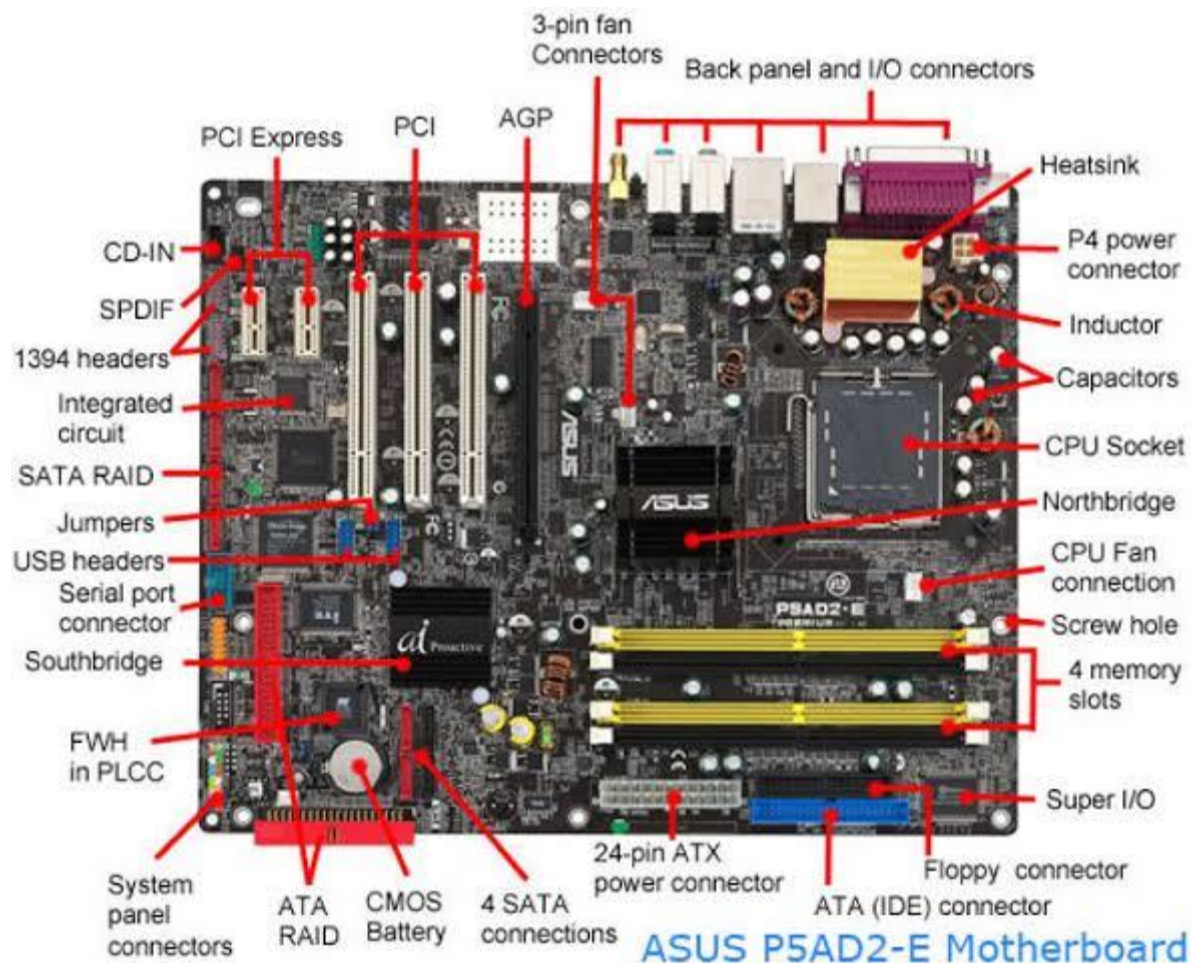
5. AM5:



Compatible CPU Families :

AMD Ryzen 7000, 8000G and 9000 series

4. Photograph or diagram the motherboard with labels.



ASUS P5AD2-E Motherboard

ComputerHope.com

Task .3 RAM & Storage Installation:

1. Safely remove the RAM module, noting ESD precautions taken.

Remove the RAM :



- 2.Reinstall the RAM module correctly and confirm it is seated properly.

Reinstall RAM:



3.Remove and reinstall the storage device, noting the connection type used (SATA/NVMe).

Remove SSD :



Reinstall SSD :



connection type used

M.2 NVMe SSD: Fits directly into the M.2 PCIe/NVMe slot; no cable is normally required.

4. Confirm both components are detected correctly after reinstallation.

Result: The RAM module and SATA storage device were safely removed and reinstalled. Both components were correctly seated/connected and detected successfully after the procedure.

Task 4: Full Desktop PC Assembly:

1. Assemble the PC using the provided motherboard, CPU, RAM, storage, GPU, and PSU.

1. the Motherboard Install

Put the motherboard inside the case and tighten the screws.

2. Install the CPU and Cooler

Put the CPU in its socket and attach the CPU cooler.

3. Install the RAM

Put the RAM into the correct slots until the clips lock.

4. Install the HDD/SSD

Put the storage drive in place and connect the data and power cables.

5. Install the GPU

Put the graphics card into the PCIe slot and tighten the screw. Connect its power cable if needed.

6. Connect SMPS Cables

Connect the 24-pin motherboard cable, CPU power cable, storage power cable, and GPU power cable if needed.

7. Connect Front-Panel Cables

Connect the Power button, USB, Audio, and Reset cables.

8. Check Everything

Check that all parts and cables are connected properly.

9. Close the Case

Put the side panel back and tighten the screws.

11. Turn On and Test

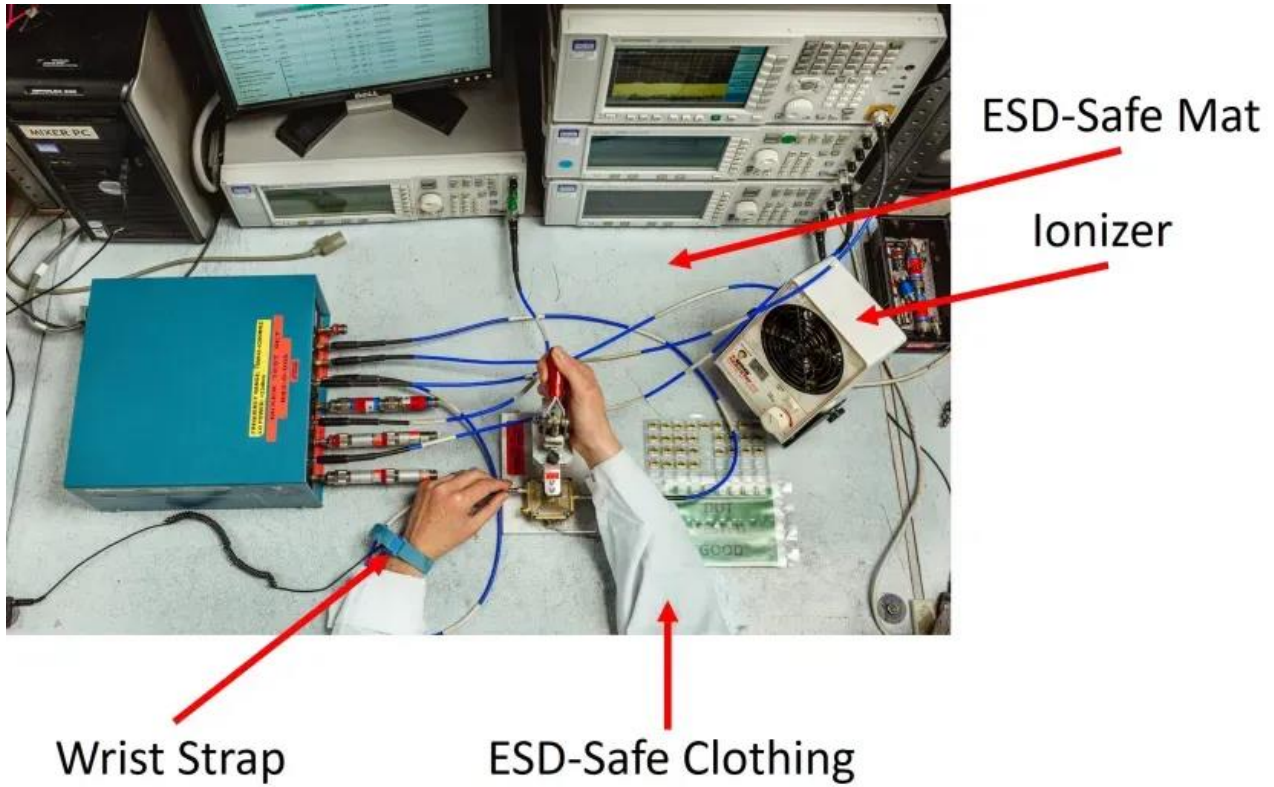
Connect the power cable, monitor, keyboard, and mouse. Turn on the PC and check that it starts normally.

2. Follow correct installation sequence and ESD/safety precautions throughout.

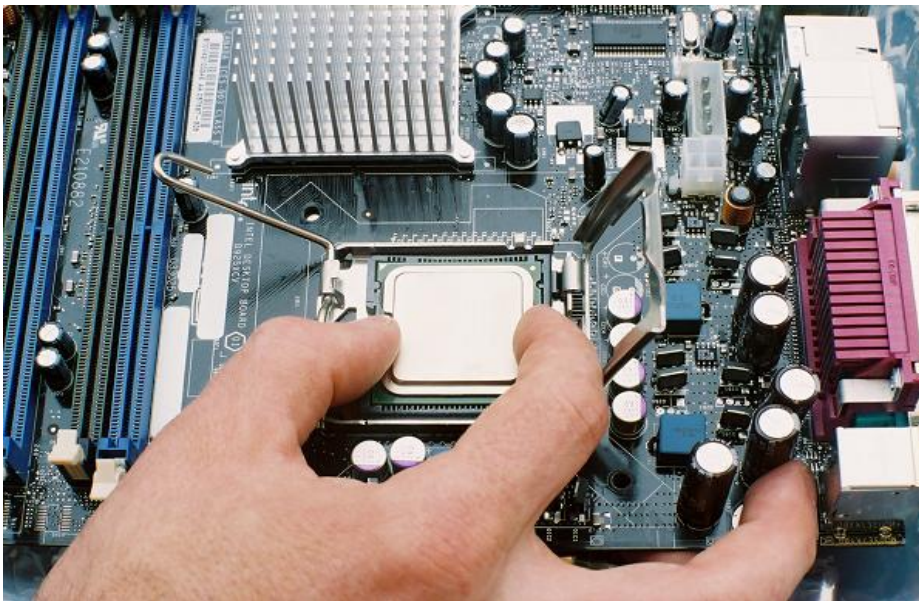
1. Switch off and unplug the PSU before starting work.
2. Work on a clean, dry, non-carpeted surface.
3. Use an anti-static wrist strap or regularly touch the grounded metal PC case to prevent ESD.
4. Handle all components by their edges and avoid touching gold contacts, pins, and connectors.
5. Install the CPU into the motherboard first, ensuring the alignment mark is correct.
6. Install the CPU cooler and connect it to the CPU_FAN header.
7. Install the RAM into the correct slots and make sure the retaining clips lock.
8. Install the M.2 NVMe SSD or mount the SATA storage device and connect its cables.
9. Install the motherboard into the case using the correct standoffs and screws.
10. Install the PSU and connect the 24-pin motherboard and CPU power cables.
11. Install the GPU into the PCIe slot and connect its PCIe power cable if required.
12. Connect the front-panel switches, USB, audio, and case-fan cables.
13. Check that there are no loose screws, damaged cables, or incorrectly connected components.
14. Reconnect power and perform a POST test.

3. Document each step with labeled photos or diagrams.

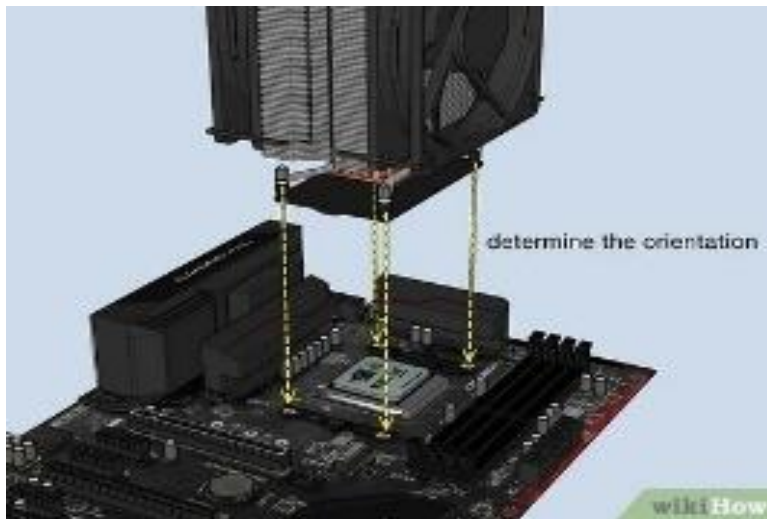
1. ESD Safety Setup



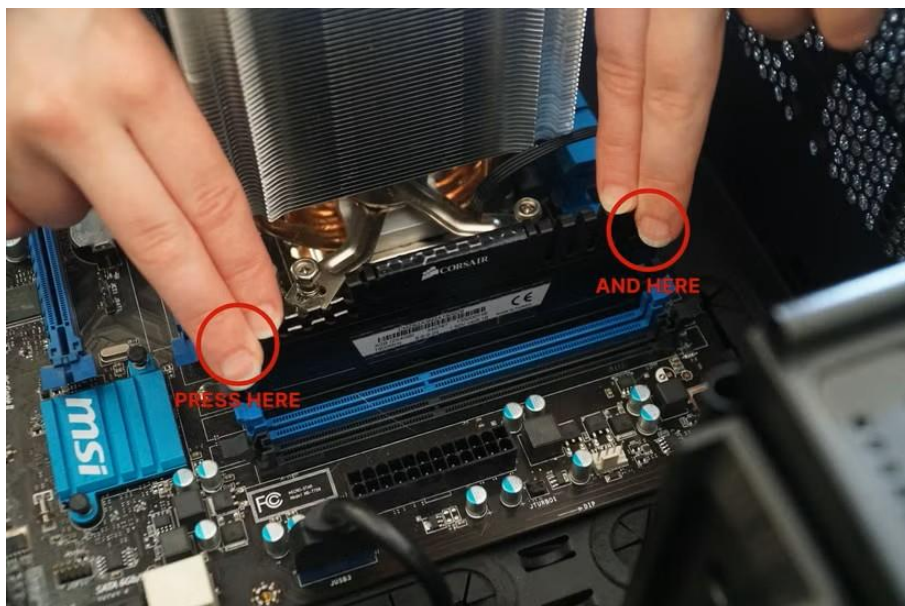
2. CPU Installation:



3. CPU Cooler Installation



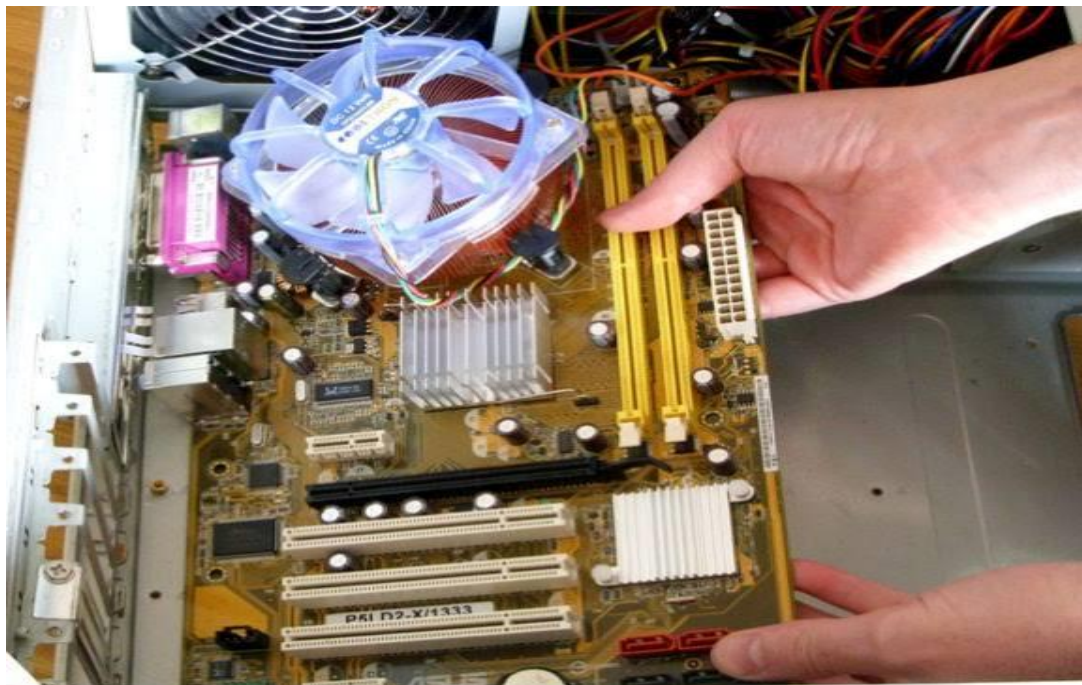
4. RAM Installation



5. Storage Installation



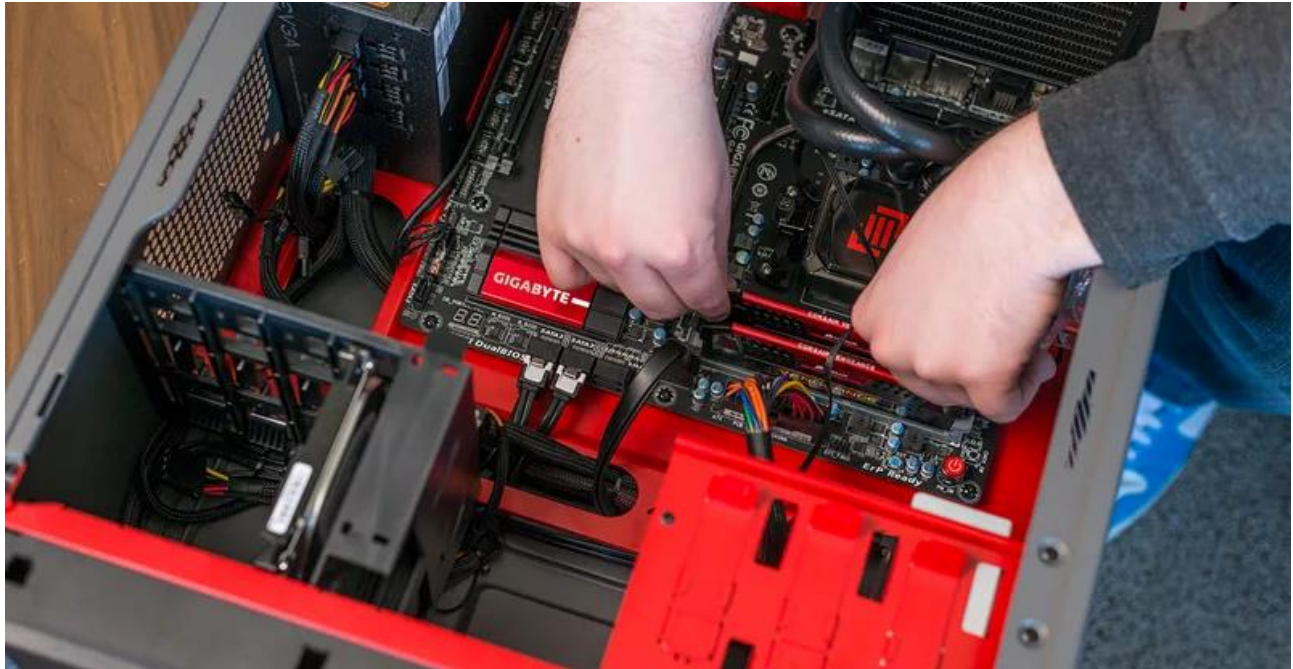
6. Motherboard Installation



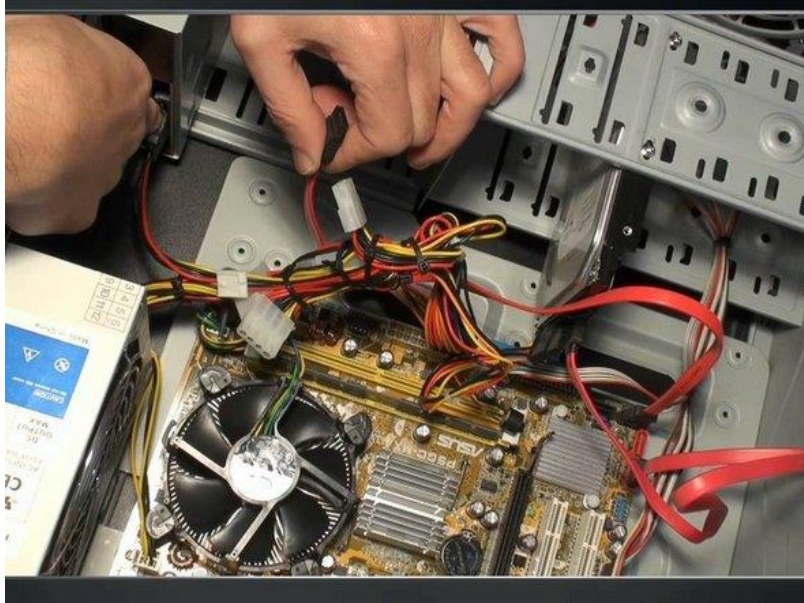
7. PSU Connections



8. GPU Installation



9. Cable Connections



10. Final Assembly



11.POST TEST .

PC powered on and BIOS/UEFI screen showing successful detection

4.Power on the system and confirm it passes POST successfully.

Power-On and POST Verification

1. Connected the monitor, keyboard, and power cable.
2. Checked that the PSU switch was set to ON (I).
3. Pressed the PC power button.
4. Observed that the fans started spinning and the system powered on.
5. Checked the motherboard POST/debug indicators for any error lights or beep codes.
6. Confirmed that the monitor displayed the BIOS/UEFI screen.
7. Checked that the CPU, RAM, and storage device were detected correctly.
8. Confirmed that the system continued to the operating system without displaying a hardware error.

Result: The PC powered on successfully and passed POST. The main hardware components were detected correctly, and the system was ready for normal operation.

Section C — Mini Capstone Project

1. Identify the three hardware components you would check first when a newly assembled PC fails to boot, and explain why.

1. PSU and Power Connections

- Check the PSU is switched on and working.
- Make sure the 24-pin motherboard and 4/8-pin CPU power cables are connected.
- A power problem can stop the PC from turning on at all.

2. RAM

- Check that the RAM is fully inserted and the clips are locked.
- Try one RAM stick at a time if necessary.
- Incorrectly seated or faulty RAM can cause POST failure or no display.

3. CPU and CPU Socket

- Check that the CPU is installed correctly and is compatible with the motherboard.
- Check the CPU cooler and CPU_FAN connection.
- An incompatible or incorrectly installed CPU can prevent the computer from completing POST.

2.Explain how motherboard–CPU compatibility affects system functionality, referencing the specific system you assembled or inspected.

Motherboard–CPU compatibility is important because the CPU must match the motherboard's socket, chipset, and supported CPU generation. The motherboard BIOS must also support the installed CPU.

For the workstation I assembled, I checked that the CPU socket matched the motherboard socket before installation. I also verified that the motherboard supported the CPU model and its BIOS version.

If an incompatible CPU is installed, the computer may fail to power on properly, fail POST, show no display, or fail to recognize the CPU. Even if the CPU physically fits the socket, it may not work if the motherboard BIOS or chipset does not support it.

3. Describe how you would verify that RAM and storage devices are installed correctly, including what BIOS/POST should confirm.

1. RAM Verification

- Turn off and unplug the PC.
- Check that the RAM notch is aligned correctly with the motherboard slot.
- Press the RAM firmly into the slot until the retaining clips lock.
- Power on the PC and check for any POST beep codes or RAM/DRAM debug LED.
- Enter BIOS/UEFI and check that the correct RAM capacity is detected.
- For example, if 16 GB RAM is installed, BIOS/UEFI should show approximately 16 GB.

2. Storage Verification

- Check that the storage device is securely installed.
- For a SATA SSD, confirm both the SATA data cable and SATA power cable are connected.
- For an NVMe SSD, confirm it is correctly inserted into the M.2 slot and secured with the screw.
- Enter BIOS/UEFI and check that the storage drive is listed.
- After booting into the operating system, confirm the drive is visible and accessible.

3. BIOS/POST Confirmation

A successful POST should complete without RAM or storage-related errors. BIOS/UEFI should correctly identify the installed RAM capacity and storage device. If either component is missing, I would power off the PC and reseat the component and check its connections.

4. Produce a hardware-installation checklist of at least three steps to prevent similar failures in future deployments.

1. Check compatibility before installation – Confirm the motherboard supports the CPU, RAM type/speed, storage interface, and GPU.
2. Follow ESD precautions – Switch off and unplug the PC, use an anti-static wrist strap, and handle components by their edges.
3. Install components correctly – Seat the CPU, RAM, GPU, and storage firmly and ensure all retaining clips/screws are secure.
4. Check power and data connections – Verify the 24-pin motherboard, CPU power, GPU power, SATA data, and SATA power cables are properly connected.
5. Perform a POST/BIOS check – Power on the PC and confirm the CPU, RAM, storage, and other hardware are detected correctly.
6. Inspect before final deployment – Check for loose cables, incorrect connections, damaged components, or unused screws inside the case.

5. Provide a technical justification for using SSDs instead of HDDs in modern office workstations, tying it back to the lab's use case.

SSD vs HDD – Simple Explanation

- SSD is faster than HDD, so the computer starts quickly.
- Programs like Microsoft Word, Excel, and web browsers open faster.
- Files can be opened and saved more quickly.
- SSDs have no moving parts, so they are more reliable and less likely to be damaged by movement.
- SSDs are quiet because they do not have spinning disks.
- SSDs use less power than HDDs.

For the ABC Solutions Lab

The lab computers will mainly be used for Office, email, web browsing, and spreadsheets. An SSD will make these computers start faster and work more smoothly.

Conclusion: SSD is a better choice than HDD for modern office computers because it is faster, quieter, and more reliable.

Section D — AI-Augmented Learning

STEP 1 · BUILD WITH AI Use an AI tool of your choice (ChatGPT, Claude, Copilot, etc.) to help you produce a step-by-step troubleshooting checklist that:

1. Guides a junior technician through diagnosing a desktop PC that fails to boot immediately after assembly, in the correct priority order

For a desktop PC that fails to boot immediately after assembly, guide the junior technician in this priority order:

1. Safety first – Turn off the PSU and unplug the power cable. Use ESD precautions.
2. Check the main power – Make sure the PSU switch is ON and the power cable is properly connected.
3. Check motherboard power – Confirm the 24-pin ATX and 4/8-pin CPU power connectors are firmly connected.
4. Check front-panel connections – Verify the Power SW connector is connected to the correct motherboard pins.
5. Check RAM – Remove and reseat the RAM. Make sure it is installed in the correct slot.
6. Check CPU and cooler – Confirm the CPU is correctly seated and the cooler is properly installed.
7. Check GPU/display – Reseat the graphics card and make sure its PCIe power cable is connected. Check that the monitor cable is connected to the correct port.
8. Check storage and other cables – Make sure SATA/M.2 drives and other components are installed correctly.
9. Try a minimal boot – Disconnect unnecessary drives and expansion devices. Boot with only motherboard, CPU, one RAM stick, PSU, and GPU if required.

10. Check POST/diagnostic signs – Look for beep codes, motherboard LEDs, or POST error messages.
11. Clear CMOS – If the system still does not POST, reset the BIOS/CMOS according to the motherboard instructions.
12. Test components individually – If necessary, test RAM, PSU, GPU, and other components with known-good parts.

Priority: Power → motherboard connections → RAM → CPU/GPU → other devices → minimal boot → POST diagnostics → component testing.

2. Covers at minimum the motherboard, RAM, storage, and PSU as possible failure points.

1. Motherboard:

Possible Problem: Power connector loose or component installed incorrectly.

What to Check: Check 24-pin and CPU power connectors, front-panel cables, and motherboard seating.

RAM:

Possible Problem: RAM not seated properly or incompatible.

What to Check: Remove and reseal RAM; try one RAM stick at a time.

Storage:

Possible Problem: SSD/HDD not connected or detected.

What to Check: Check SATA/data and power cables, or reseal the M.2 SSD.

PSU:

Possible Problem: PSU switched off, faulty, or insufficient power.

What to Check: Check PSU switch, power cable, 24-pin/CPU connectors, and PSU output.

3. Is presented as a clear checklist or simple decision flow a technician could follow on-site.

Desktop PC No-Boot Checklist

1. Make the system safe

- Shut down and unplug the PC.
- Use an anti-static wrist strap or other ESD precautions.
- Do not force any connector.

2. Check the PSU

- Confirm the PSU switch is ON.
- Check the power cable and wall outlet.
- Make sure the 24-pin motherboard connector is secure.
- Make sure the 4/8-pin CPU power connector is secure.
- Check GPU power connectors if a dedicated GPU is installed.

3. Check the motherboard

- Confirm the motherboard is mounted correctly.
- Check for loose screws or possible shorts.
- Check the front-panel Power SW connection.
- Look for motherboard diagnostic LEDs or error codes.

4. Check the RAM

- Remove and reseal the RAM.
- Check that the RAM clips lock into place.
- Try booting with one RAM stick.
- Try the recommended RAM slot from the motherboard manual.

5. Check the CPU

- Confirm the CPU is correctly seated.

- Check for damaged socket pins where applicable.
- Make sure the CPU cooler is properly installed.
- Confirm the CPU fan is connected to the CPU_FAN header.

6. Check the GPU

- Make sure the graphics card is fully seated in the PCIe slot.
- Connect its required PCIe power cables.
- Check that the monitor cable is connected correctly.
- If supported, test using integrated graphics.

7. Check storage

- For SATA drives, check both SATA data and power cables.
- For M.2 SSDs, make sure the drive is fully seated and secured.
- Check whether the storage drive appears in BIOS.

8. Try a minimal boot

- Disconnect unnecessary USB devices and expansion cards.
- Leave only the motherboard, CPU, cooler, one RAM stick, PSU, and required graphics.
- Try starting the PC again.

9. Check POST

- Listen for beep codes.
- Check motherboard diagnostic LEDs.
- Look for POST error messages.
- Enter BIOS if the display appears.

10. If BIOS opens

- Check whether RAM is detected.

- Check whether the SSD/HDD is detected.
- Check CPU temperature and basic hardware information.
- Check the correct boot device.

11. If there is still no POST

- Clear CMOS according to the motherboard instructions.
- Reseat RAM and GPU again.
- Test with known-good RAM or PSU if available.

12. Final decision

- No power → investigate PSU, power cable, and motherboard power.
- Power but no POST → investigate RAM, CPU, motherboard, and GPU.
- POST works but no operating system → investigate storage and boot settings.
- BIOS detects everything but Windows/Linux does not start → investigate the boot drive or operating system.

4. Is applied to the ABC Solutions scenario from Section C to show which step would have caught the reported issue

ABC Solutions Scenario – Applying the Troubleshooting Checklist

In the ABC Solutions scenario, the reported problem was that the newly assembled desktop failed to boot.

The step that would most likely have caught the issue was:

Step 2 – Check the PSU and power connections.

The technician should have checked that the PSU was switched on and that the 24-pin motherboard power connector and 4/8-pin CPU power connector were firmly connected.

If those were correct, the next important check would be Step 4 – Check the RAM, by reseating the RAM and testing one stick at a time.

How the checklist helps:

No boot → Check PSU → Check motherboard power → Check RAM → Check CPU/GPU → Minimal boot → Check POST.

This follows the correct troubleshooting order and helps the technician find the fault before replacing unnecessary components.

STEP 2 · TEST & DEBUG (WITHOUT AI)

Then, working without AI, review the checklist and find at least one step that is missing, incorrectly ordered, or unsafe (for example, skipping ESD precautions or checking software before physical connections). Fix it yourself.

I reviewed the hardware installation checklist and found that ESD precautions were missing.

Problem: The checklist started with installing components without first protecting the components from static electricity.

Fix: I added this step at the beginning:

1. Turn off and unplug the PC, then use an anti-static wrist strap or touch a grounded metal part before handling components.
2. Check that all components are correctly installed and cables are firmly connected.
3. Only after checking the physical connections, turn on the PC and check POST/BIOS.
4. If the PC does not start, check the power connections, RAM, CPU, and other hardware before checking software.

1. The exact prompt(s) you gave the AI tool.

"Create a hardware-installation checklist of at least three steps to prevent similar failures in future deployments."

2. The AI's original checklist and your corrected version.

AI's Original Checklist:

- Check that all components are compatible.
- Install the CPU, RAM, storage, and other components correctly.
- Connect all power and data cables.
- Turn on the PC and check that it starts correctly.

My Corrected Checklist:

- Turn off and unplug the PC and use ESD protection before handling components.
- Check component compatibility and install the CPU, RAM, storage, and other components correctly.
- Connect all power and data cables securely and check that nothing is loose.
- Turn on the PC and check POST/BIOS for errors before checking software.

3. A 3–4 line note explaining what you changed and why the AI's version needed it.

I added ESD protection as the first step because it was missing from the AI checklist. I also made sure the physical connections are checked before turning on the PC. The AI version was generally correct, but it did not clearly include ESD safety or the correct troubleshooting order.